



Lecture Notes

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Unit No.:	2	Unit Name:	Numerical Methods

Unit 2: Numerical Methods

Numerical methods are techniques used to obtain approximate, computable solutions to mathematical problems that are difficult or impossible to solve exactly by analytical means. This unit introduces methods for solving systems of linear equations (both direct and iterative), the classification of numerical errors, methods for locating the roots of algebraic and transcendental equations, and interpolation techniques for estimating unknown function values from a set of known data points.

Learning Objectives

- Solve systems of linear equations using the Gauss Elimination and Gauss-Jordan methods.
- Apply the Jacobi and Gauss-Seidel iterative methods to solve linear systems and verify convergence.
- Distinguish between absolute, relative, and percentage error.
- Locate roots of algebraic and transcendental equations using the Bisection, Regula-Falsi, and Secant methods.
- Apply Newton's forward and backward difference interpolation formulas to estimate intermediate values of a function.
- Use Lagrange's interpolation formula and Newton's divided difference formula to interpolate unequally spaced data.

1. Solution of Linear System of Equations

Direct methods solve a linear system in a fixed number of steps by systematically reducing the system to a simpler equivalent form using elementary row operations.

1.1 Gauss Elimination Method

A system of linear equations can be written compactly as an augmented matrix, in which the coefficients of the variables and the right-hand-side constants are arranged together. The matrix containing only the coefficients (without the constants) is called the coefficient matrix.

For example, the system:

$$3x + 4y = 7, \quad 4x - 2y = 5$$

is written as the augmented matrix $[3 \ 4 \mid 7; 4 \ -2 \mid 5]$, with coefficient matrix $[3 \ 4; 4 \ -2]$.

Similarly, a three-by-three system such as:

$$3x - y - z = 0, \quad x + y = 5, \quad 2x - 3z = 2$$

is represented by the augmented matrix $[3 \ -1 \ -1 \mid 0; 1 \ 1 \ 0 \mid 5; 2 \ 0 \ -3 \mid 2]$, with coefficient matrix $[3 \ -1 \ -1; 1 \ 1 \ 0; 2 \ 0 \ -3]$.

In the Gauss Elimination method, elementary row operations ($R_i \rightarrow R_i - k \cdot R_j$) are applied to the augmented matrix to reduce the coefficient matrix to upper-triangular form (all entries below the leading diagonal become zero). The resulting triangular system is then solved by back-substitution, starting from the last equation.

Example: Solve the system $x + y + z = 2$, $x + 2y + 3z = 5$, $2x + 3y + 4z = 11$.

Solution:

The augmented matrix $[1 \ 1 \ 1 \mid 2; 1 \ 2 \ 3 \mid 5; 2 \ 3 \ 4 \mid 11]$ is reduced step by step:

- $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 2R_1$ gives $[1 \ 1 \ 1 \mid 2; 0 \ 1 \ 2 \mid 3; 0 \ 1 \ 2 \mid 7]$.
- $R_3 \rightarrow R_3 - R_2$ gives $[1 \ 1 \ 1 \mid 2; 0 \ 1 \ 2 \mid 3; 0 \ 0 \ 0 \mid 4]$.
- $R_1 \rightarrow R_1 - R_2$ gives $[1 \ 0 \ -1 \mid -1; 0 \ 1 \ 2 \mid 3; 0 \ 0 \ 0 \mid 4]$.

The last row, $0 = 4$, is a contradiction, so this particular system is inconsistent and has no solution. (The example illustrates how row reduction reveals inconsistency; for a consistent system the triangular form is solved by back-substitution from the last non-zero row upward.)

Practice Problems

1. Solve $x - y = -6$, $6x - 2y = 0$ using the Gauss Elimination method.
2. Solve $x + y + z = 6$, $3x + y + 6z = 10$, $2x + 4y - z = 1$ using the Gauss Elimination method.
3. Solve $2x - y + 3z = 9$, $x + y + z = 6$, $x - y + z = 2$ using the Gauss Elimination method.

1.2 Gauss-Jordan Method

The Gauss-Jordan method extends Gauss Elimination by continuing the row reduction until the coefficient matrix becomes the identity matrix (reduced row-echelon form), so that the solution can be read off directly without back-substitution.

Example: Solve $x - 2y = 4$, $-5y + z = -9$, $4x - 3z = -10$ by the Gauss-Jordan method.

Solution:

Writing the augmented matrix $[1 \ -2 \ 0 \mid -4; 0 \ -5 \ 1 \mid -9; 4 \ 0 \ -3 \mid -10]$ and applying row operations:

- $R_3 \rightarrow R_3 - 4R_1$ gives $[1 \ -2 \ 0 \mid -4; 0 \ -5 \ 1 \mid -9; 0 \ 8 \ -3 \mid 6]$.
- $R_2 \rightarrow (-1/5)R_2$ gives $[1 \ -2 \ 0 \mid -4; 0 \ 1 \ -1/5 \mid 9/5; 0 \ 8 \ -3 \mid 6]$.
- $R_1 \rightarrow R_1 + 2R_2$ and $R_3 \rightarrow R_3 - 8R_2$ gives $[1 \ 0 \ -2/5 \mid -2/5; 0 \ 1 \ -1/5 \mid 9/5; 0 \ 0 \ -7/5 \mid -42/5]$.
- $R_3 \rightarrow (-5/7)R_3$ gives $[1 \ 0 \ -2/5 \mid -2/5; 0 \ 1 \ -1/5 \mid 9/5; 0 \ 0 \ 1 \mid 6]$.
- $R_1 \rightarrow R_1 + (2/5)R_3$ and $R_2 \rightarrow R_2 + (1/5)R_3$ gives $[1 \ 0 \ 0 \mid 2; 0 \ 1 \ 0 \mid 3; 0 \ 0 \ 1 \mid 6]$.

The solution read directly from the identity matrix is $X = 2$, $Y = 3$, $Z = 6$.

Practice Problems

1. Solve $x - y = -6$, $6x - 2y = 0$ using the Gauss-Jordan method.
2. Solve $x + y + z = 5$, $2x + 3y + 5z = 8$, $4x + 5z = 2$ using the Gauss-Jordan method.
3. Solve $2x - y + 3z = 9$, $x + y + z = 6$, $x - y + z = 2$ using the Gauss-Jordan method.

2. Iterative Methods: Jacobi and Gauss-Seidel Methods

Unlike direct methods, iterative methods start from an initial approximation and generate a sequence of improved approximations that converge to the true solution. They are especially useful for large, sparse systems where direct elimination is computationally expensive. A sufficient condition commonly used to check convergence is diagonal dominance: in each equation, the magnitude of the diagonal coefficient must exceed the sum of the magnitudes of the other coefficients in that row.

2.1 Jacobi Method

For a system of linear equations $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$, $a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$, ..., $a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$, each variable is isolated on its own diagonal term:

$$x_1 = (1/a_{11})[b_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n]$$

$$x_2 = (1/a_{22})[b_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n], \dots, x_n = (1/a_{nn})[b_n - a_{n1}x_1 - \dots - a_{n,n-1}x_{n-1}]$$

In the Jacobi method, every variable in an iteration is updated using only the values obtained from the previous iteration (all updates happen simultaneously).

Example: Solve $26x_1 + 2x_2 + 2x_3 = 12.6$, $3x_1 + 27x_2 + x_3 = -14.3$, $2x_1 + 3x_2 + 17x_3 = 6.0$ by the Jacobi method, correct to three decimal places.

Solution:

Convergence check: $26 > 2 + 2$, $27 > 3 + 1$, $17 > 2 + 3$ — the system is diagonally dominant, so the Jacobi iterations will converge. Taking the initial approximation $x_1(0) = x_2(0) = x_3(0) = 0$:

kth iteration	0	1	2	3	4	5
x_1	0.000	0.48462	0.49821	0.50006	0.50000	0.50001
x_2	0.000	-0.52963	-0.59655	-0.59941	-0.59999	-0.60000
x_3	0.000	0.35294	0.38939	0.39960	0.39989	0.40000

After the fifth iteration the successive differences are all smaller than 0.0002, so the process is stopped. The solution, correct to three decimal places, is $x_1 = 0.500$, $x_2 = -0.600$, $x_3 = 0.400$.

Practice Problems

1. Solve $5x_1 - 2x_2 + 3x_3 = -1$, $-3x_1 + 9x_2 + x_3 = 2$, $2x_1 - x_2 - 7x_3 = 3$ using the Jacobi method, correct to three decimal places.
2. Solve $x_1 - 5x_2 = -4$, $7x_1 - x_2 = 6$ using the Jacobi method.

A further exercise is to solve $45x_1 + 2x_2 + 3x_3 = 58$, $-3x_1 + 22x_2 + 2x_3 = 47$, $5x_1 + x_2 + 20x_3 = 67$ by the Jacobi method (correct to three decimal places) — first verify diagonal dominance: $45 > 2 + 3$, $22 > -3 + 2$, $20 > 5 + 1$.

2.2 Gauss-Seidel Method

The Gauss-Seidel method modifies the Jacobi method so that, within each iteration, a variable is updated using the most recently computed values of the other variables (rather than waiting for the next full iteration). This typically produces faster convergence than the Jacobi method.

Example: Consider $10x_1 - x_2 + 2x_3 = 6$, $-x_1 + 11x_2 - x_3 + 3x_4 = 25$, $2x_1 - x_2 + 10x_3 - x_4 = -11$, $3x_2 - x_3 + 8x_4 = 15$.

Solution:

Isolating each variable on its diagonal term gives:

$$x_1 = x_2/10 - x_3/5 + 3/5, \quad x_2 = x_1/11 + x_3/11 - 3x_4/11 + 25/11$$

$$x_3 = -x_1/5 + x_2/10 + x_4/10 - 11/10, \quad x_4 = -3x_2/8 + x_3/8 + 15/8$$

Starting from the initial approximation (0, 0, 0, 0), using each variable's most recently updated value as soon as it is available:

$$x_1 = 3/5 = 0.6, \quad x_2 = 2.3272, \quad x_3 = -0.9873, \quad x_4 = 0.8789 \text{ (first iteration)}$$

x1	x2	x3	x4
0.6	2.32727	-0.987273	0.878864
1.03018	2.03694	-1.01446	0.984341
1.00659	2.00356	-1.00253	0.998351
1.00086	2.0003	-1.00031	0.99985

The iterations converge rapidly to the exact solution (1, 2, -1, 1).

Example: Solve $45x_1 + 2x_2 + 3x_3 = 58$, $-3x_1 + 22x_2 + 2x_3 = 47$, $5x_1 + x_2 + 20x_3 = 67$ by the Gauss-Seidel method, correct to three decimal places.

Solution:

Convergence check: $45 > 2 + 3$, $22 > -3 + 2$, $20 > 5 + 1$ — the system is diagonally dominant.

kth iteration	0	1	2	3	4
x1	0.000	1.28889	0.99198	0.99958	1.0000
x2	0.000	2.31212	2.00689	1.99979	1.99999
x3	0.000	2.91217	3.00166	3.00012	3.00000

The solution, correct to three decimal places, is $x_1 = 1.000$, $x_2 = 2.000$, $x_3 = 3.000$.

3. Types of Errors: Absolute, Relative, and Percentage Error

Since numerical methods produce approximate rather than exact answers, it is essential to quantify how far an approximation deviates from the true value. Three related measures are used:

$$\text{Absolute Error} = | \text{True Value} - \text{Measured (Approximate) Value} |$$

Absolute error expresses the magnitude of the difference between a measurement/approximation and the true value, in the same units as the quantity being measured.

$$\text{Relative Error} = \text{Absolute Error} / \text{True Value}$$

Relative error expresses the size of the error in proportion to the true value, which makes it possible to compare the accuracy of approximations of different magnitudes.

$$\text{Percentage Error} = \text{Relative Error} \times 100\%$$

Percentage error expresses the relative error as a percentage of the true (or accepted) value, and is the form most commonly quoted when reporting the accuracy of a numerical result.

4. Solution of Algebraic and Transcendental Equations

Many equations $f(x) = 0$ arising in practice — particularly transcendental equations involving trigonometric, exponential, or logarithmic terms — cannot be solved exactly. The following iterative methods locate an approximate root by successive refinement of an initial estimate or bracketing interval.

4.1 Bisection Method

For a continuous function $f(x)$, the Bisection method proceeds as follows:

- Find two points a and b such that $a < b$ and $f(a) \cdot f(b) < 0$ (the function changes sign, so a root lies between a and b).
- Compute the midpoint $t = (a + b)/2$.
- If $f(t) = 0$, then t is the root; otherwise, determine in which half-interval the sign change occurs: if $f(t) \cdot f(a) < 0$, the root lies between a and t (set $b = t$); if $f(t) \cdot f(b) < 0$, the root lies between t and b (set $a = t$).
- Repeat the above steps until $f(t)$ is sufficiently close to zero or the interval $[a, b]$ is sufficiently small.

Example: Determine the root of $x^2 - 3 = 0$ for $x \in [1, 2]$.

Solution:

Let $f(x) = x^2 - 3$. Since $f(1) = -2 < 0$ and $f(2) = 1 > 0$, and f is continuous, a root lies in $[1, 2]$. Successive bisection gives the following iterations:

Iteration	a	b	t	f(a)	f(b)	f(t)
1	1	2	1.5	-2	1	-0.75

Iteration	a	b	t	f(a)	f(b)	f(t)
2	1.5	2	1.75	-0.75	1	0.062
3	1.5	1.75	1.625	-0.75	0.0625	-0.359
4	1.625	1.75	1.6875	-0.359	0.0625	-0.1523
5	1.6875	1.75	1.7188	-0.1523	0.0625	-0.0457
6	1.7188	1.75	1.7344	-0.0457	0.0625	0.0081
7	1.7188	1.7344	1.7266	-0.0457	0.0081	-0.0189

At the seventh iteration, the final interval is [1.7266, 1.7344]. Hence, the approximated solution is 1.7344 (the true value is $\sqrt{3} \approx 1.7321$).

Practice Problems

1. Use the bisection method to find the root of $f(x) = x^2 - 5$ in the interval $[2, 3]$, up to 4 decimal places.
2. Apply the bisection method to solve $f(x) = \cos(x) - x$ in the interval $[0, 1]$, up to 3 decimal places.
3. Use the bisection method to find the root of $f(x) = x^3 - 2x - 5$ in the interval $[2, 3]$, up to 5 decimal places.
4. Solve $f(x) = x^2 - 2x - 3$ for a root in the interval $[1, 4]$ using the bisection method.
5. Use the bisection method to approximate the root of $f(x) = e^x - 3$ in the interval $[0, 2]$, to 3 decimal places.

4.2 Regula-Falsi (False Position) Method

The Regula-Falsi method is similar to bisection in that it brackets the root between two points of opposite sign, but instead of bisecting the interval it uses linear interpolation between $(a, f(a))$ and $(b, f(b))$ to estimate the root, which usually converges faster than bisection:

- Choose two initial points a and b such that $f(a) \cdot f(b) < 0$.
- Compute the point c where the line joining $(a, f(a))$ and $(b, f(b))$ crosses the x -axis, using $c = [a \cdot f(b) - b \cdot f(a)] / [f(b) - f(a)]$.
- Determine $f(c)$. If $f(c) \cdot f(a) < 0$, the root lies between a and c (set $b = c$); if $f(c) \cdot f(b) < 0$, the root lies between b and c (set $a = c$).
- Repeat until $|f(c)|$ is smaller than a predefined tolerance, or $[a, b]$ is sufficiently small.

Example: Find a root of $2e^x \sin x = 3$, correct to three decimal places, using three iterations of the false position method.

Solution:

Let $f(x) = 2e^x \sin x - 3$. Testing values: $f(0) = -3$, $f(0.5) \approx -1.420$, $f(1) \approx 1.574$. Since $f(0.5) < 0$ and $f(1) > 0$, the root lies in $[0.5, 1]$.

Iteration	a	b	f(a)	f(b)	xr	f(xr)
1	0.5	1.0	-1.420	1.574	0.73714	-0.191
2	0.73714	1.0	-0.191	1.574	0.76558	-0.021
3	0.76558	1.0	-0.021	1.574	0.76842	-0.003

The root after three iterations is approximately 0.768 (to three decimal places).

4.3 Newton-Raphson Method

The Newton-Raphson method uses the tangent line to $f(x)$ at the current approximation to obtain the next, generally better, approximation. Given a differentiable function $f(x)$ and an initial guess x_0 , the iteration formula is:

$$x_{n+1} = x_n - f(x_n) / f'(x_n), \text{ for } n = 0, 1, 2, \dots$$

Geometrically, x_{n+1} is the point where the tangent to the curve at $(x_n, f(x_n))$ crosses the x-axis. The method converges very rapidly (quadratically) near a simple root, provided the initial guess x_0 is reasonably close to the root and $f'(x_n) \neq 0$ at each step.

- Choose an initial approximation x_0 close to the expected root.
- Compute $f(x_n)$ and $f'(x_n)$ at the current approximation.
- Update the approximation using $x_{n+1} = x_n - f(x_n)/f'(x_n)$.
- Repeat until $|x_{n+1} - x_n|$ (or $|f(x_{n+1})|$) is smaller than a predefined tolerance.

Example: Find the root of $f(x) = x^3 - 2x - 5 = 0$, correct to four decimal places, with $x_0 = 2$.

Solution:

Here $f(x) = x^3 - 2x - 5$ and $f'(x) = 3x^2 - 2$.

Iteration	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}
1	2.000000	-1.000000	10.000000	2.100000
2	2.100000	0.061000	11.230000	2.094568
3	2.094568	0.000186	11.161647	2.094551
4	2.094551	0.000000	11.161438	2.094551

The root, correct to four decimal places, is 2.0946 — the same root located by the Secant method below, but reached in fewer iterations since Newton-Raphson uses the actual derivative rather than a secant-line approximation to it.

Practice Problems

1. Find $\sqrt{7}$ (the positive root of $f(x) = x^2 - 7 = 0$), correct to five decimal places, using the Newton-Raphson method with $x_0 = 3$. (Hint: $f'(x) = 2x$.)
2. Find the root of $f(x) = e^{-x} - x = 0$, correct to five decimal places, using the Newton-Raphson method with $x_0 = 0.5$. (Hint: $f'(x) = -e^{-x} - 1$.)
3. Find the root of $f(x) = \cos(x) - x = 0$, correct to four decimal places, using the Newton-Raphson method with $x_0 = 0.5$. (Hint: $f'(x) = -\sin(x) - 1$.)

4.4 Secant Method

The Secant method also uses linear interpolation between two points, but unlike Regula-Falsi it does not require the two points to bracket the root — it simply uses the two most recent approximations. Given two initial guesses x_0 and x_1 , the next approximation is computed as:

$$x_{(n+1)} = x_n - f(x_n) \cdot (x_n - x_{(n-1)}) / (f(x_n) - f(x_{(n-1)})), \text{ for } n = 1, 2, 3, \dots$$

Example: Find the root of $f(x) = x^3 - 2x - 5 = 0$, correct to four decimal places, with $x_0 = 2$, $x_1 = 3$.

Solution:

Iteration	$x_{(n-1)}$	x_n	$f(x_{(n-1)})$	$f(x_n)$	$x_{(n+1)}$
1	2.000000	3.000000	-1.000000	16.000000	2.058824
2	3.000000	2.058824	16.000000	-1.390715	2.096558
3	2.058824	2.096558	-1.390715	-0.147204	2.094518
4	2.096558	2.094518	-0.147204	0.003068	2.094554
5	2.094518	2.094554	0.003068	-0.000007	2.094551
6	2.094554	2.094551	-0.000007	0.000000	2.094551

The root, correct to four decimal places, is 2.0946.

Practice Problems

1. Find $\sqrt{7}$ (the positive root of $f(x) = x^2 - 7 = 0$), correct to five decimal places, using $x_0 = 2$, $x_1 = 3$ as initial guesses. (Worked answer: 2.64575.)
2. Find the root of $f(x) = e^{-x} - x = 0$, correct to five decimal places, using $x_0 = 0$, $x_1 = 1$ as initial guesses.

5. Interpolation

Interpolation is the technique of estimating the value of a function for an intermediate value of the independent variable, lying within the range of a given set of known values. (Estimating the value of the function outside the given range is called extrapolation.) The following methods construct an interpolating polynomial from a table of known (x, y) values.

5.1 Newton's Forward Difference Interpolation

This method is used when the data points are equally spaced and the value to be interpolated lies near the beginning of the table. The forward differences are defined recursively — $\Delta y_0 = y_1 - y_0$, $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$, and so on — and arranged in a forward difference table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
x_0	y_0			
x_1	y_1	Δy_0		
x_2	y_2	Δy_1	$\Delta^2 y_0$	
x_3	y_3	Δy_2	$\Delta^2 y_1$	$\Delta^3 y_0$

Newton's Gregory Forward Interpolation Formula:

$$f(a + hu) = f(a) + u \cdot \Delta f(a) + [u(u-1)/2!] \cdot \Delta^2 f(a) + \dots + [u(u-1) \dots (u-n+1)/n!] \cdot \Delta^n f(a)$$

Here h is the interval of difference (spacing between successive x -values), a is the first tabulated value of x , and $u = (x - a)/h$. This formula is particularly useful for interpolating values of $f(x)$ near the beginning of the table.

Example: Given the population figures below, find the population in 1895 using Newton's forward difference formula.

x	1891	1901	1911	1921	1931
y (population)	46	66	81	93	101

Solution:

The forward difference table (Δy , $\Delta^2 y$, $\Delta^3 y$, $\Delta^4 y$) is constructed from the y -values: $\Delta y = 20, 15, 12, 8$; $\Delta^2 y = -5, -3, -4$; $\Delta^3 y = 2, -1$; $\Delta^4 y = -3$. Here $h = 1901 - 1891 = 10$, and for $x = 1895$, $u = (1895 - 1891)/10 = 0.4$. Substituting into Newton's forward formula:

$$y(1895) = 46 + 0.4(20) + [0.4(0.4-1)/2](-5) + [0.4(0.4-1)(0.4-2)/6](2) + [0.4(0.4-1)(0.4-2)(0.4-3)/24](-3)$$

$$y(1895) = 46 + 8 + 0.6 + 0.128 + 0.1248 = 54.8528$$

Hence, the estimated population in 1895 is 54.8528 (thousand).

5.2 Newton's Backward Difference Interpolation

This method is used when the value to be interpolated lies near the end of the table. The backward differences are defined as $\nabla y_r = y_r - y_{r-1}$, and Newton's Gregory Backward Interpolation Formula is:

$$f(a + nh + uh) = f(a + nh) + u \cdot \nabla f(a + nh) + [u(u+1)/2!] \cdot \nabla^2 f(a + nh) + \dots + [u(u+1) \dots (u+n-1)/n!] \cdot \nabla^n f(a + nh)$$

Here h is the interval of difference, a_n (the last tabulated value of x) is used as the base point, and $u = (x - a_n)/h$.

Example: Using the same population data (x : 1891, 1901, 1911, 1921, 1931; y : 46, 66, 81, 93, 101), estimate the population in 1925.

Solution:

Taking the last value $a_n = 1931$ as the base, the backward difference table gives $\nabla y = 8$ ($=101-93$), $\nabla^2 y = -4$, $\nabla^3 y = -1$, $\nabla^4 y = -3$ (built upward from the bottom of the table). With $h = 10$ and $u = (1925 - 1931)/10 = -0.6$, substitution into the backward formula gives the estimated population in 1925 as 96.8368 (thousand).

A second example: given $x = 50, 100, 150, 200, 250$ with $y = 618, 724, 805, 906, 1032$, the backward difference table ($\nabla y = 106, 81, 101, 126$; $\nabla^2 y = -25, 20, 25$; $\nabla^3 y = 45, 5$; $\nabla^4 y = -40$) is used to estimate $y(300)$, taking $a_n = 250$ as the base and $u = (300 - 250)/50 = 1$; the result is $y(300) = 1148$.

Practice Problems

1. From the data x : 0, 0.1, 0.2, 0.3, 0.4 and y : 1, 1.095, 1.179, 1.251, 1.310, construct a difference table and estimate y when $x = 0.7$.
2. From the data x : 0, 1, 2, 3 and y : 1, 2, 5, 10, construct a difference table and estimate the value of x for which $y = 7.0$.

5.3 Lagrange's Interpolation

Lagrange's interpolation formula constructs an n th-degree polynomial that passes exactly through a given set of (x, y) points, without requiring the x -values to be equally spaced. For two points (x_0, y_0) and (x_1, y_1) , the first-order Lagrange formula is:

$$y = [(x - x_1)/(x_0 - x_1)] \cdot y_0 + [(x - x_0)/(x_1 - x_0)] \cdot y_1$$

Example: Find the value of y at $x = 2$ for the points $(1, 2)$ and $(3, 4)$.

Solution:

Here $(x_0, y_0) = (1, 2)$ and $(x_1, y_1) = (3, 4)$. At $x = 2$: $y = [(2-3)/(1-3)] \times 2 + [(2-1)/(3-1)] \times 4 = (-1/-2) \times 2 + (1/2) \times 4 = 1 + 2 = 3$. The value of y at $x = 2$ is 3.

For three points (x_0, y_0) , (x_1, y_1) , (x_2, y_2) , the second-order Lagrange formula is:

$$y = \frac{[(x-x_1)(x-x_2)/(x_0-x_1)(x_0-x_2)] \cdot y_0}{[(x-x_0)(x-x_1)/(x_2-x_0)(x_2-x_1)] \cdot y_2} + \frac{[(x-x_0)(x-x_2)/(x_1-x_0)(x_1-x_2)] \cdot y_1}{[(x-x_0)(x-x_1)/(x_2-x_0)(x_2-x_1)] \cdot y_2} + \frac{[(x-x_0)(x-x_1)/(x_2-x_0)(x_2-x_1)] \cdot y_2}{[(x-x_0)(x-x_1)/(x_2-x_0)(x_2-x_1)] \cdot y_2}$$

Example: Find the value of y at $x = 1$ for the points $(1, 6)$, $(3, 4)$, $(2, 5)$.

Solution:

Here $(x_0, y_0) = (1, 6)$, $(x_1, y_1) = (3, 4)$, $(x_2, y_2) = (2, 5)$. At $x = 1$: $y = [(1-3)(1-2)/(1-3)(1-2)] \times 6 + [(1-1)(1-2)/(3-1)(3-2)] \times 4 + [(1-1)(1-3)/(2-1)(2-3)] \times 5 = (12/2) + 0 + 0 = 6$. The value of y at $x = 1$ is 6.

A further worked example: for the points $(9, 6)$, $(3, 5)$, $(1, 12)$, the value of y at $x = 10$ works out to $y = 63/8 - 15/4 + 21/4 = 75/8 = 9.375$.

Practice Problems

1. Find the value of y at $x = 5$ for the given set of points $(9, 2), (3, 10)$.
2. Find the value of y at $x = 7$ for the given set of points $(1, 10), (2, 4), (3, 4), (5, 7)$.

5.4 Newton's Divided Difference Interpolation

Newton's divided difference formula is used when the x -values in the data are not equally spaced, so that ordinary forward/backward differences cannot be applied directly. Divided differences are defined as:

$$f[x_0, x_1] = (f_1 - f_0)/(x_1 - x_0), \quad f[x_0, x_1, x_2] = (f[x_1, x_2] - f[x_0, x_1])/(x_2 - x_0), \quad \text{and so on.}$$

Newton's divided difference interpolation formula is then:

$$f(x) = f[x_0] + (x-x_0)f[x_0, x_1] + (x-x_0)(x-x_1)f[x_0, x_1, x_2] + \dots + (x-x_0)(x-x_1)\dots(x-x_{n-1})f[x_0, x_1, \dots, x_n]$$

Example: Given the data $(1, 2), (2, 3), (4, 7), (7, 15)$, find the cubic Newton's interpolating polynomial.

Solution:

The divided difference table is:

x	$f(x)$	1st	2nd	3rd
1	2			
2	3	1		